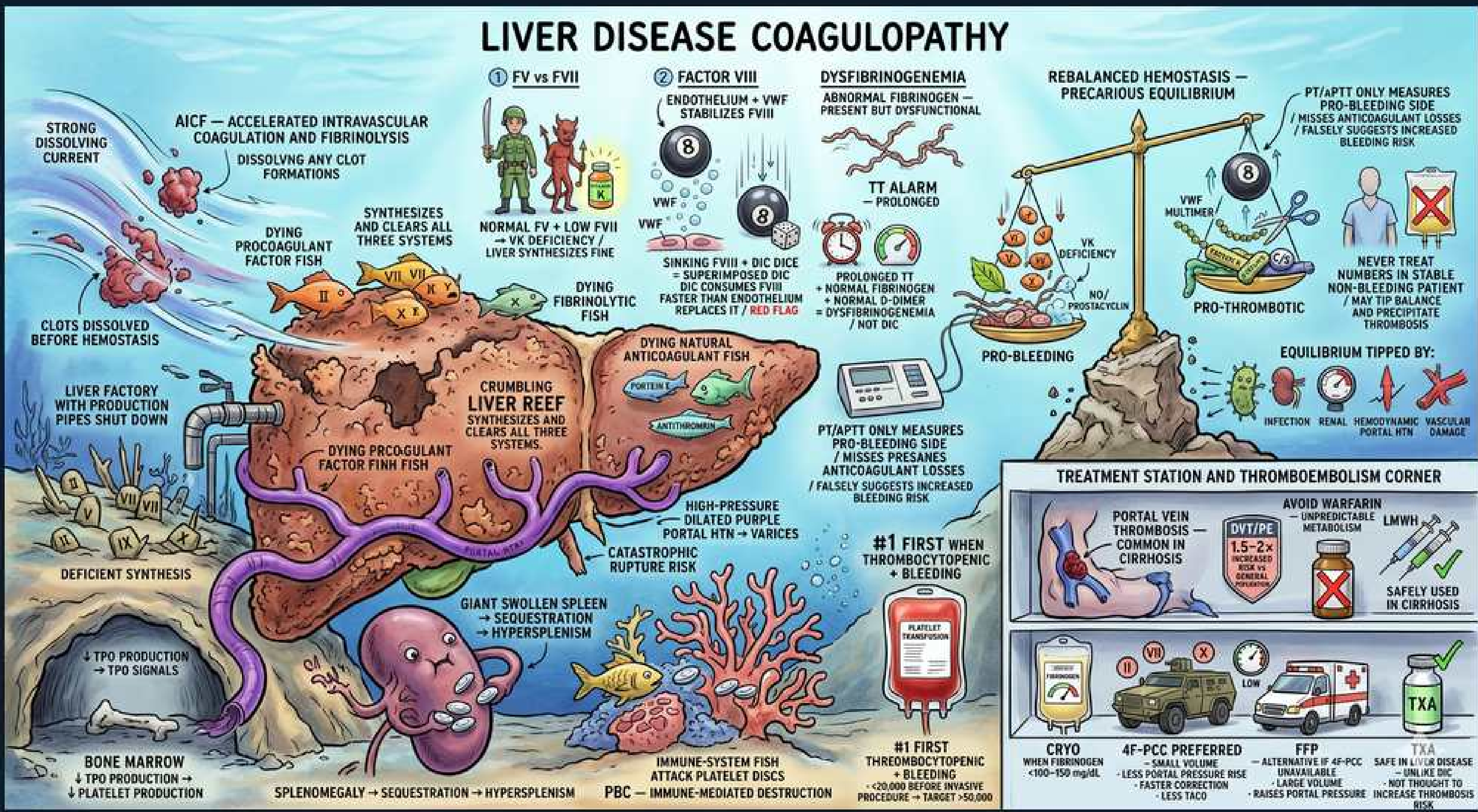
Coagulation — Liver Disease Coagulopathy



1. Liver makes most factors

WHAT YOU SEE

Center crumbling 'liver reef' synthesizing procoagulant fish

WHAT IT ENCODES

The hepatocyte is the factory for nearly ALL coagulation factors — I (fibrinogen), II, V, VII, IX, X, XI — plus the natural anticoagulants and fibrinolytic proteins. Cirrhosis/synthetic failure reduces production across the board → a complex coagulopathy. The shortest-half-life factor (VII, ~4–6 h) falls first, so PT/INR is the earliest abnormal screen and tracks synthetic function (it is part of MELD). Mnemonic: 'the crumbling reef can't build the fish.'

BOARD TRAP

Synthetic failure lowers MANY factors at once and prolongs PT before aPTT. The one striking EXCEPTION is factor VIII (see next) — remembering that exception is the whole game on the boards.

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2. Factor VIII spared (normal/high)

WHAT YOU SEE

Top-left 'FV vs FVIII, normal FV low or DIC clue, FVIII normal/high'

WHAT IT ENCODES

Factor VIII is the great EXCEPTION: it is synthesized largely by vascular ENDOTHELIUM (sinusoidal/extrahepatic), not solely by hepatocytes, AND it is an acute-phase reactant. So in liver disease FVIII is NORMAL or even HIGH despite every other factor falling. Mnemonic: 'VIII comes from the VEIN wall, not the liver.'

BOARD TRAP

A LOW factor VIII argues AGAINST pure liver disease and FOR DIC (where consumption drops everything including VIII). 'All factors low but VIII preserved' = liver; 'all factors low including VIII' = DIC.

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3. FVIII distinguishes liver from DIC

WHAT YOU SEE

Center FVIII '#1 distinguisher liver vs DIC, normal/high in liver, low in DIC'

WHAT IT ENCODES

Factor VIII level is THE single best lab to separate liver coagulopathy from DIC. Liver disease: FVIII normal/high. DIC: FVIII consumed and LOW. Supportive clues: DIC also shows markedly rising D-dimer and falling fibrinogen with schistocytes, whereas chronic liver disease tends to be more stable. Mnemonic: 'VIII high = liver lie; VIII low = DIC, oh no.'

BOARD TRAP

PT, platelets, and fibrinogen overlap heavily between liver disease and DIC — do NOT try to distinguish them on PT/platelets alone. Order factor VIII (and factor V) to differentiate.

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4. Low factor V = liver

WHAT YOU SEE

'Dying natural anticoagulant fish' / low FV note

WHAT IT ENCODES

Factor V is made by the liver but is NOT vitamin K–dependent. In hepatic synthetic failure factor V falls along with the others; in pure vitamin K deficiency factor V is NORMAL. So FV separates a hepatic synthetic problem from a vitamin K problem. Combined with FVIII, factor V triangulates the cause. Mnemonic: 'Five and Eight tell the fate — low V = liver, low VIII = DIC.'

BOARD TRAP

Vitamin K will NOT fix a low factor V — that's hepatic synthetic failure, not a carboxylation defect. If giving vitamin K corrects the PT, the problem was vitamin K deficiency (FV would have been normal all along).

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5. PT/INR prolonged

WHAT YOU SEE

Right hourglass 'PT/INR only measures pro-bleeding side, prolonged'

WHAT IT ENCODES

PT/INR is prolonged in cirrhosis and is used to grade synthetic function/severity (a MELD component). BUT the PT/INR was designed and calibrated to monitor WARFARIN — it measures only the PROCOAGULANT (pro-bleeding) factors and is blind to the simultaneous fall in natural ANTICOAGULANTS. Mnemonic: 'INR sees only half the scale.'

BOARD TRAP

A prolonged INR in cirrhosis does NOT reliably predict bleeding risk and should NOT be used to assess pre-procedure bleeding risk the way it is in a warfarin patient. Many cirrhotics with a 'high INR' are not auto-anticoagulated at all.

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6. Rebalanced hemostasis

WHAT YOU SEE

Top-right scale 'rebalanced hemostasis, precarious equilibrium'

WHAT IT ENCODES

Cirrhosis lowers BOTH procoagulants (II, V, VII, IX, X, fibrinogen) AND anticoagulants (protein C, S, antithrombin), and platelets fall while von Willebrand factor RISES. The net effect is a fragile, REBALANCED hemostasis — a new equilibrium that can tip toward either bleeding OR clotting under stress (infection, renal failure, procedures). Mnemonic: 'the scale is balanced but the pans are nearly empty.'

BOARD TRAP

An abnormal INR does NOT mean the patient is 'auto-anticoagulated' or protected from thrombosis — cirrhotics still develop portal vein thrombosis and VTE. The balance is precarious in both directions.

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7. Low protein C/S/antithrombin

WHAT YOU SEE

A center 'dying natural-anticoagulant fish' panel 'liver clears anticoagulants too' — protein C, S and antithrombin all fall, balancing the lost procoagulants

WHAT IT ENCODES

The liver also manufactures the natural ANTICOAGULANTS — protein C, protein S, and antithrombin. Their decline in cirrhosis OFFSETS the loss of procoagulant factors, which is exactly why hemostasis stays 'rebalanced' rather than purely pro-bleeding. Falling protein C/S/AT is the prothrombotic counterweight. Mnemonic: 'brakes fail at the same time as the engine.'

BOARD TRAP

The drop in natural anticoagulants is the reason cirrhotics STILL clot (portal vein thrombosis, DVT/PE) despite a long INR. Don't equate low factors with a bleeding-only state.

8. Decreased thrombopoietin

WHAT YOU SEE

Bottom-left bone-marrow scene 'TPO producer — reduced TPO signals' — the liver-made platelet hormone drops, cutting production

WHAT IT ENCODES

Thrombopoietin (TPO), the master driver of platelet production, is made by the LIVER. Hepatic dysfunction lowers TPO → reduced megakaryocyte stimulation → decreased platelet PRODUCTION, contributing to thrombocytopenia independent of the spleen. Mnemonic: 'no liver, no TPO, no platelets made.'

BOARD TRAP

Thrombocytopenia in cirrhosis is partly a PRODUCTION problem (low TPO), not entirely splenic sequestration — which is why TPO-receptor agonists (e.g., avatrombopoietin) can raise counts before procedures.

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9. Splenic sequestration

WHAT YOU SEE

A bottom-center swollen octopus-spleen 'giant swollen spleen — sequestration / hypersplenism' — portal hypertension pools the platelets

WHAT IT ENCODES

Portal hypertension → congestive SPLENOMEGALY → hypersplenism, which pools/sequesters up to ~90% of the platelet mass. Combined with low TPO, this produces the typical mild-to-moderate thrombocytopenia of cirrhosis. Mnemonic: 'the big spleen octopus grabs the platelets.'

BOARD TRAP

Cirrhotic thrombocytopenia is mainly HYPERSPLENISM + low TPO, NOT immune destruction. Don't reflexively diagnose ITP or transfuse platelets to a fixed number in a non-bleeding cirrhotic — sequestered platelets are still functionally available and vWF is high.

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10. Dysfibrinogenemia

WHAT YOU SEE

Center 'dysfibrinogenemia, abnormal fibrinogen present but dysfunctional, TT prolonged'

WHAT IT ENCODES

The diseased liver produces structurally ABNORMAL fibrinogen (acquired dysfibrinogenemia) — fibrinogen is present in normal/near-normal amount by antigen but doesn't polymerize properly. This prolongs the thrombin time (TT) and reptilase time. Mnemonic: 'the bricks are there but they're warped.'

BOARD TRAP

Fibrinogen ANTIGEN/level can be normal while fibrinogen FUNCTION (TT, clottable fibrinogen) is impaired — measuring only the antigen misses dysfibrinogenemia. A long TT in liver disease points here.

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11. Vitamin K may help

WHAT YOU SEE

A right note flagging the cholestasis/vitamin-K-deficiency component — impaired bile-dependent vitamin K absorption adds a correctable layer

WHAT IT ENCODES

Cholestasis in liver disease impairs bile-dependent absorption of fat-soluble vitamin K, adding a superimposed (and CORRECTABLE) vitamin K–deficiency component. A therapeutic trial of vitamin K can partially shorten the PT if deficiency is contributing. Mnemonic: 'before blaming the liver, give the K.'

BOARD TRAP

If the PT FULLY corrects with vitamin K, the problem was deficiency, not synthetic failure. If it does NOT correct, it is true hepatic synthetic failure (factor V will be low) — and vitamin K won't fix that.

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12. Portal vein thrombosis

WHAT YOU SEE

Bottom-right 'portal vein thrombosis common in cirrhosis, LMWH'

WHAT IT ENCODES

Because natural anticoagulants fall and flow is sluggish, cirrhotics are PROthrombotic enough to develop portal vein thrombosis (PVT) and systemic VTE. Anticoagulation (LMWH, or a DOAC in compensated disease) may be indicated DESPITE an abnormal baseline INR, and can even improve survival/recanalization in PVT. Mnemonic: 'cirrhosis clots the portal vein.'

BOARD TRAP

Do NOT assume cirrhotics are 'auto-anticoagulated' and protected from clotting — that myth leads to missed/undertreated thrombosis. The high INR is not a contraindication to needed anticoagulation.

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13. Cryo/4F-PCC/FFP for bleeding

WHAT YOU SEE

Treatment station bottom 'cryo, 4F-PCC preferred, FFP, TXA'

WHAT IT ENCODES

Treat only ACTIVE bleeding or genuinely high-risk procedures, not numbers: cryoprecipitate (or fibrinogen concentrate) to keep fibrinogen >100–120 mg/dL, 4F-PCC (low volume, preferred) or FFP to replace factors, platelets if profoundly low, and antifibrinolytics (TXA) for mucosal bleeding. Viscoelastic testing (TEG/ROTEM) guides targeted product use. Mnemonic: 'fix the patient, not the lab.'

BOARD TRAP

Do NOT transfuse FFP/platelets to 'normalize' the INR in a STABLE, non-bleeding cirrhotic — it doesn't reduce bleeding, raises portal pressure (volume), and risks TACO/thrombosis. The INR target paradigm of warfarin does not apply here.
